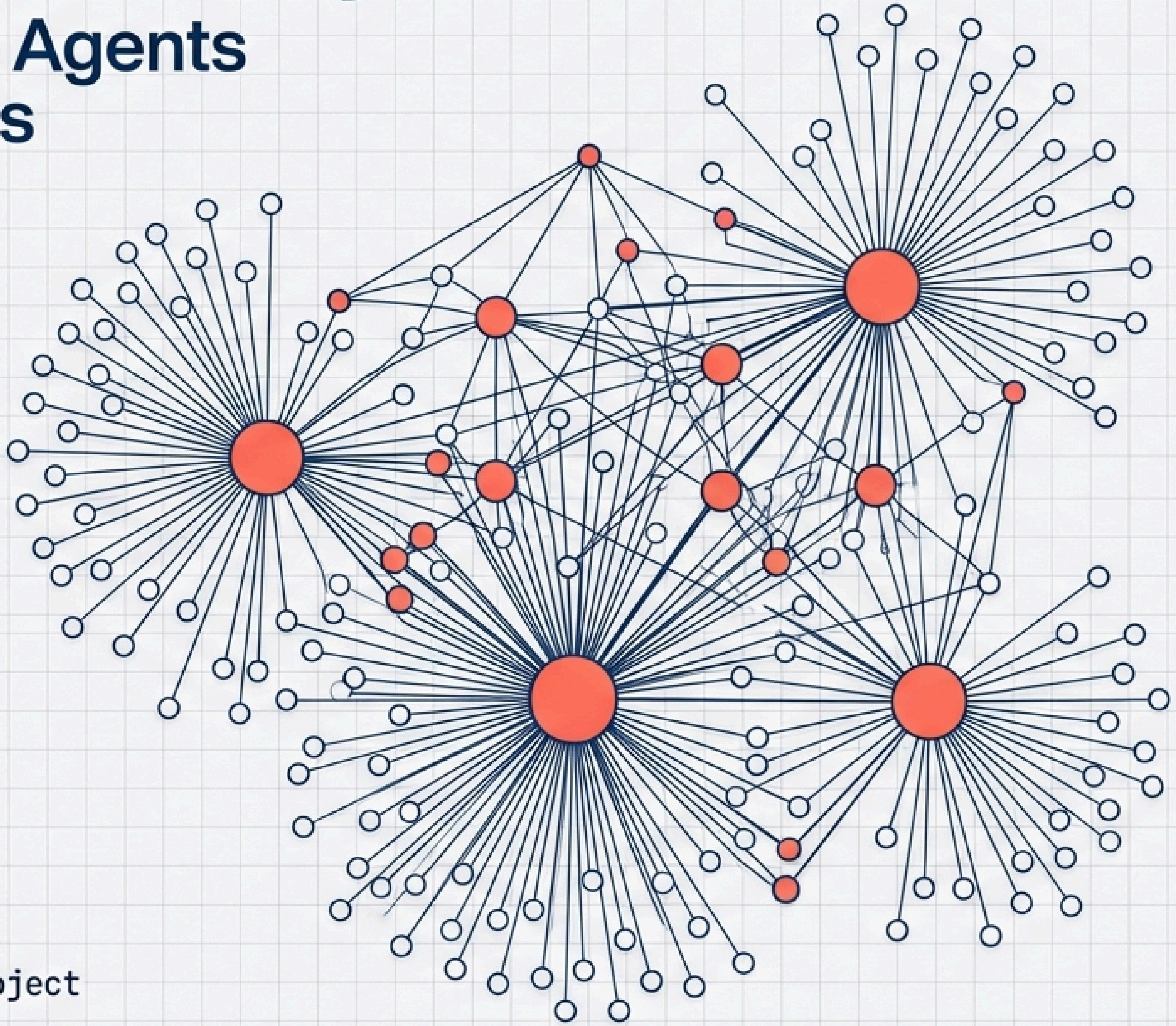
Network Security Games

From Strategic Agents to Logic Solvers



Algorithmic Game Theory Project
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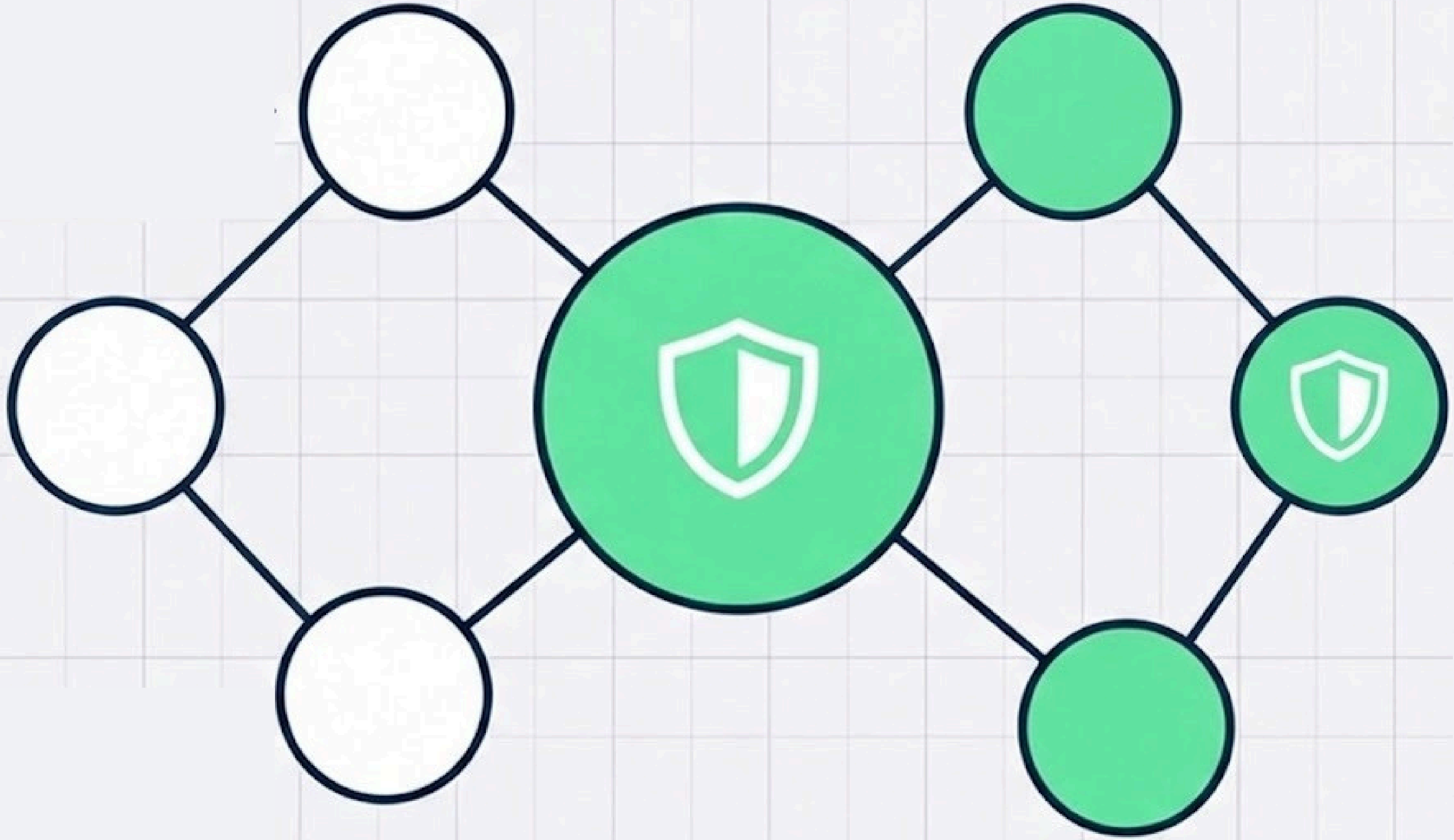
The Objective



Goal: Secure Graph G=(V,E)

The Rule: A set S is valid if every node is in S OR has a neighbor in S.

Equivalence: Minimal Vertex Cover (NP-Hard)

Task 1: Strategic Game Modeling

Investing Node (Secure)

Net Payoff = 6

Free-riding Node (Covered)

Net Payoff = 10

Game Type: Strategic Substitutes (Local Public Good)

Parameters: Value α=10, Cost c=4 | **Goal:** Find Pure Nash Equilibria (PNE)

Reaching Equilibrium

Best Response Dynamics

Strictly improving updates.

Sequential logic prevents cycles.

Fictitious Play

History-based beliefs.

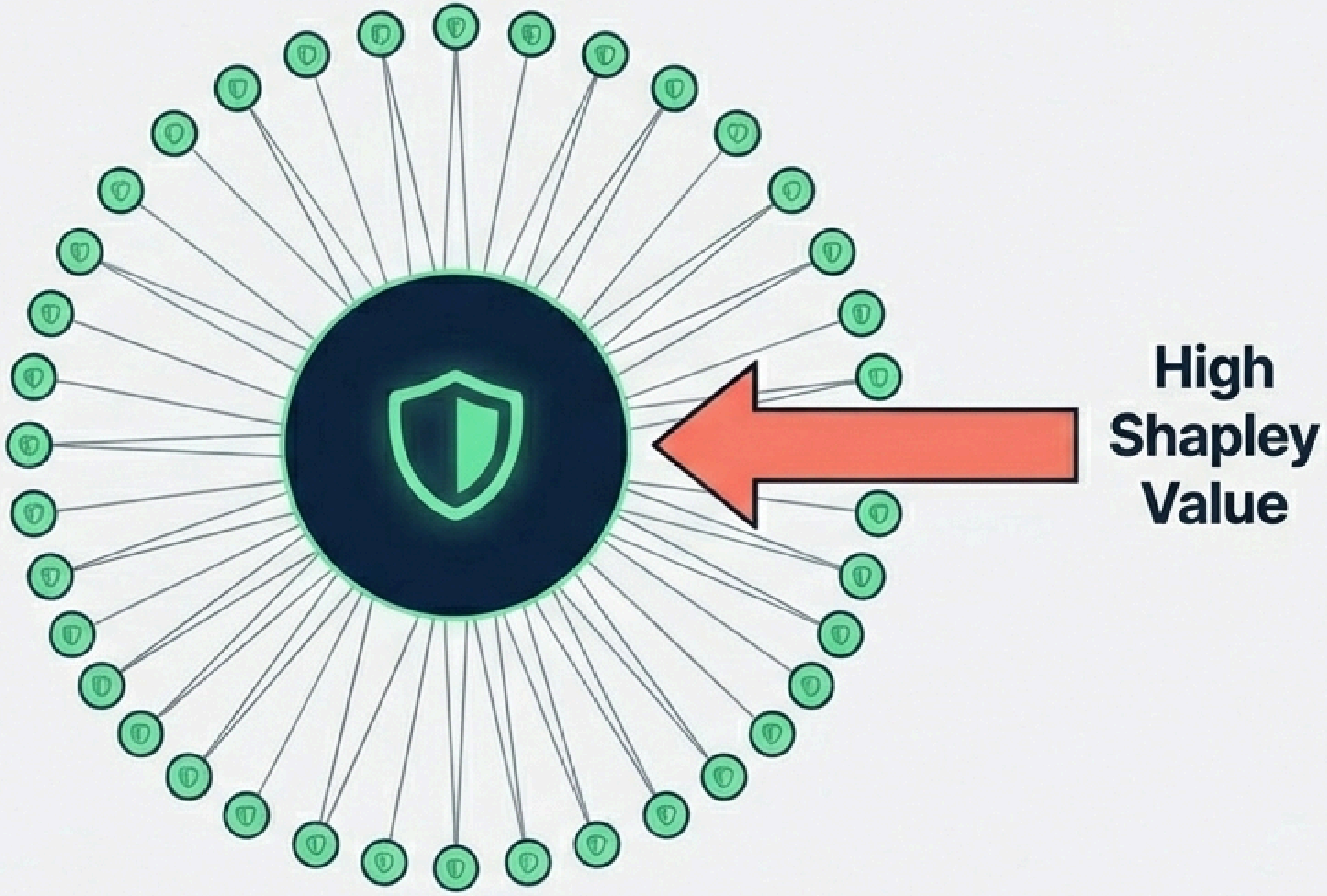
Deterministic threshold response.

Regret Matching

Minimizes cumulative regret.

Probabilistic selection.

Task 2: Coalitional Analysis



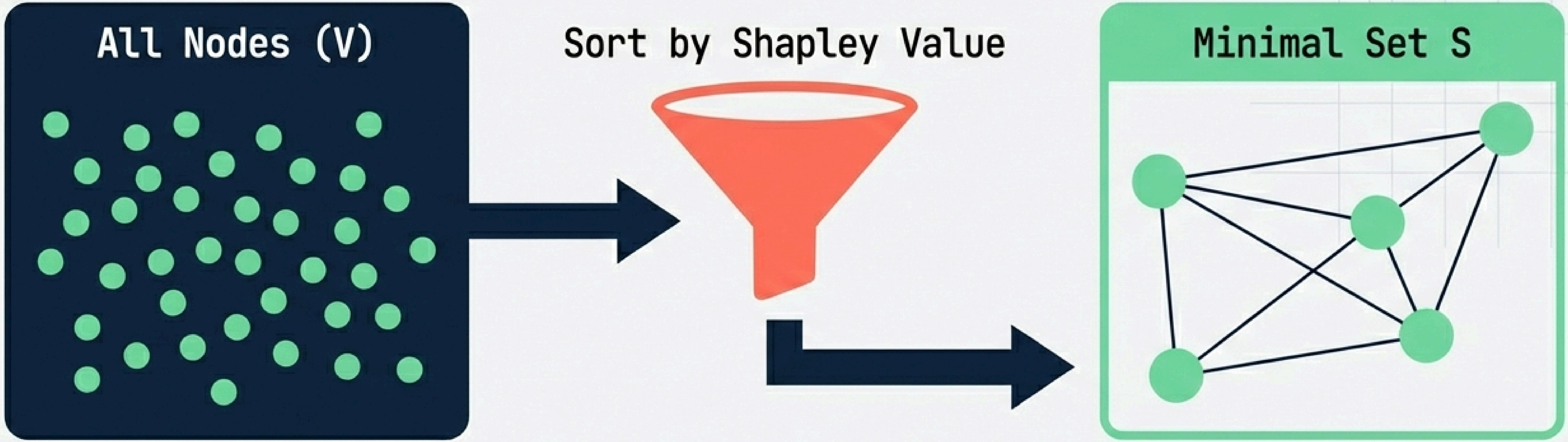
Metric: Shapley Value (Average Marginal Contribution)

Challenge: N! Complexity (Intractable)

Solution: Monte Carlo Approximation

Contribution = +1 if node is the *last* missing neighbor.

The Reverse Greedy Heuristic

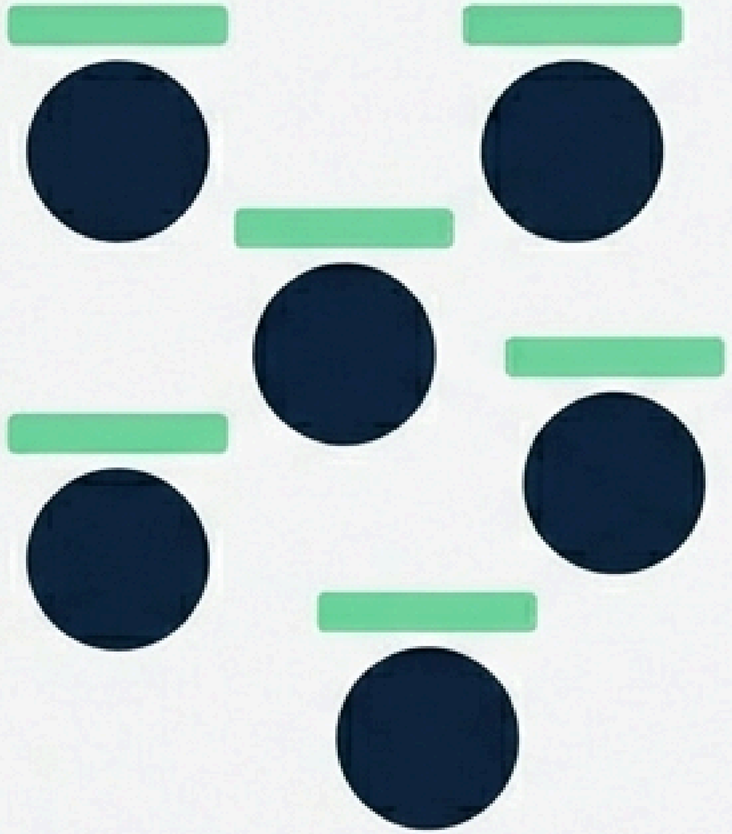


Logic: Start Full -> Prune Sequentially

Constraint: Remove node ‘i’ ONLY IF neighbors remain safe.

Task 3: Market Allocation

Secure Nodes (Budget B)



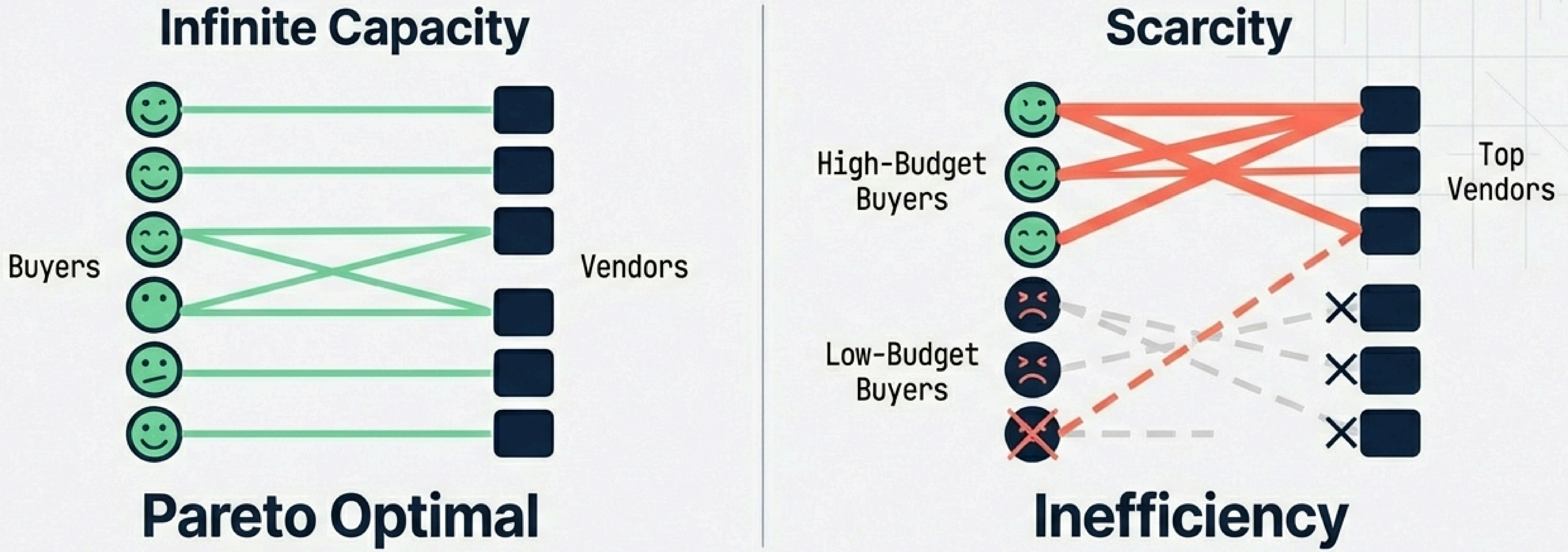
Providers (Quality Q, Capacity K)

Utility = (Quality × 10) + (Budget - Price)

$$\text{Utility} = (\text{Quality} \times 10) + (\text{Budget} - \text{Price})$$

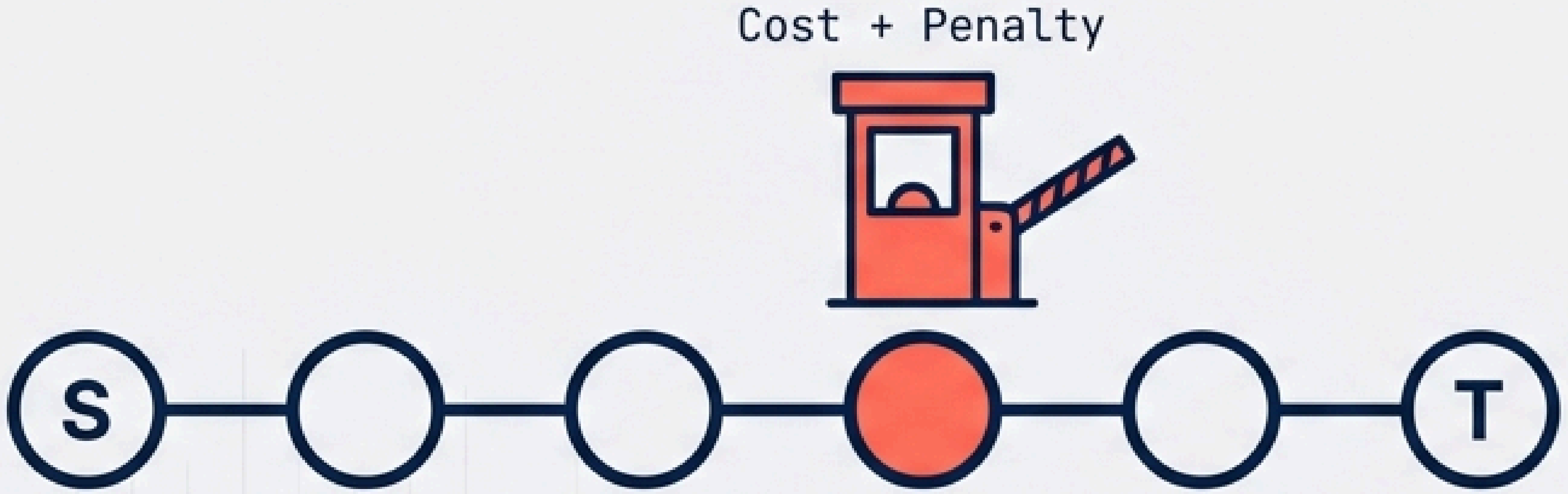
Goal: Maximize Social Welfare

Efficiency vs. Crowding Out



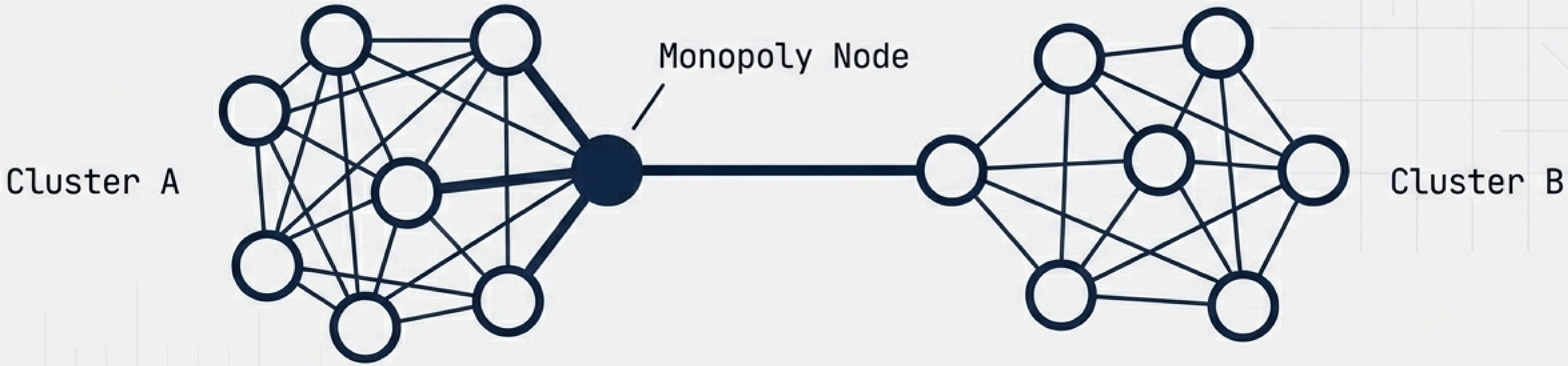
Key Insight: Infinite Supply: Local Greedy -> Global Optimum
Scarcity: Wealthier buyers crowd out others

Task 4: VCG Path Auction



- **Goal:** Procure secure path s -> t
- **Mechanism:** VCG (Truthful)
- **Cost Function:** Bid + Insecurity Penalty (Externality)

The VCG Payment Rule



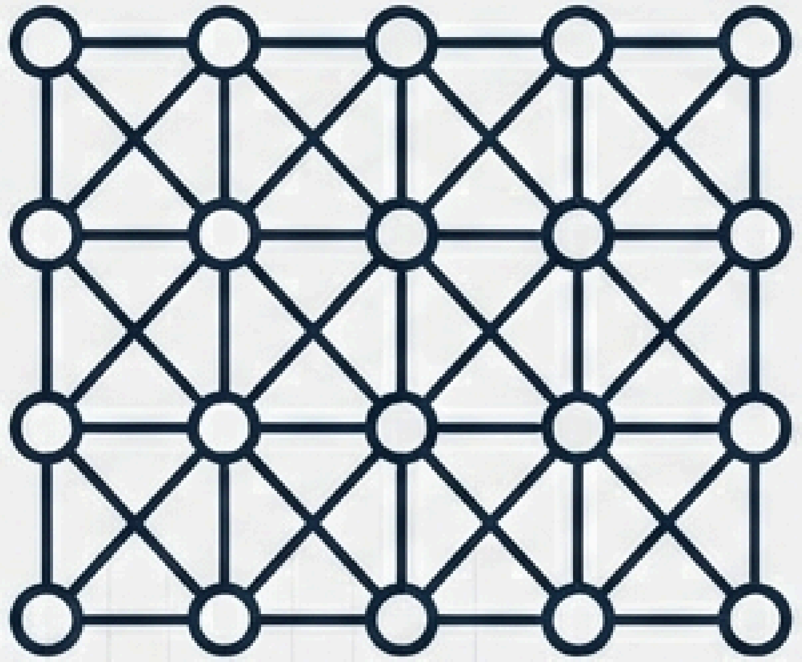
Payment = Externality

$$P_i = SP(G \text{ without } i) - (SP(G) - w_i)$$

Bridge nodes extract maximum payments.

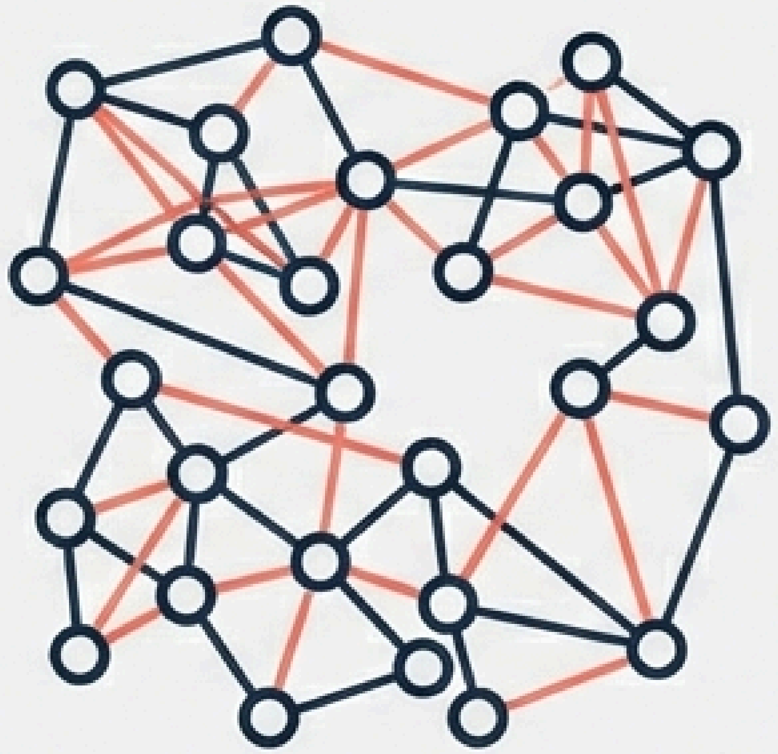
Topology Dictates Destiny

Regular Graph



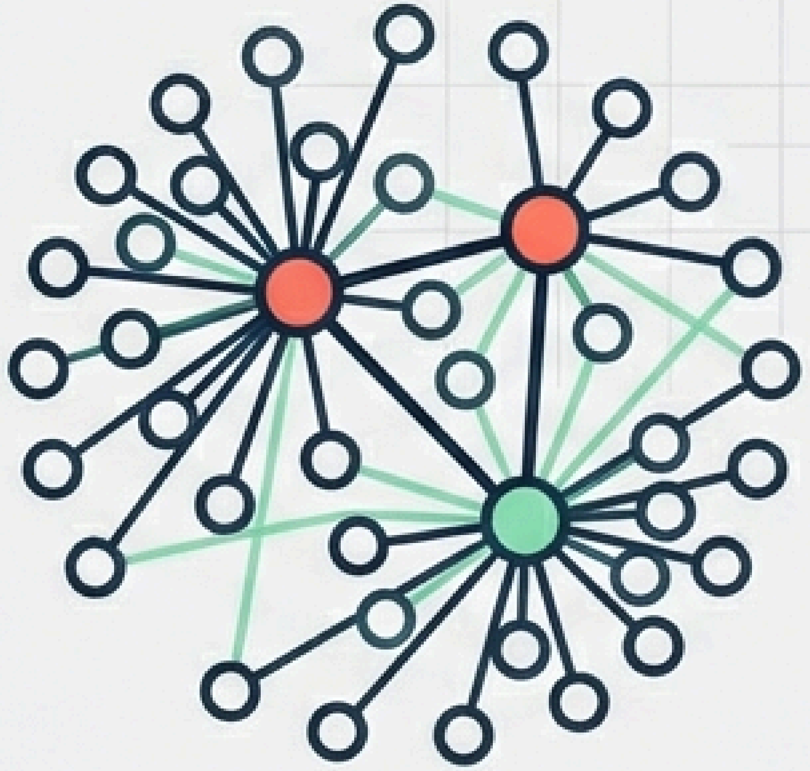
Distributed Responsibility

Erdős-Rényi



Probabilistic Uniformity

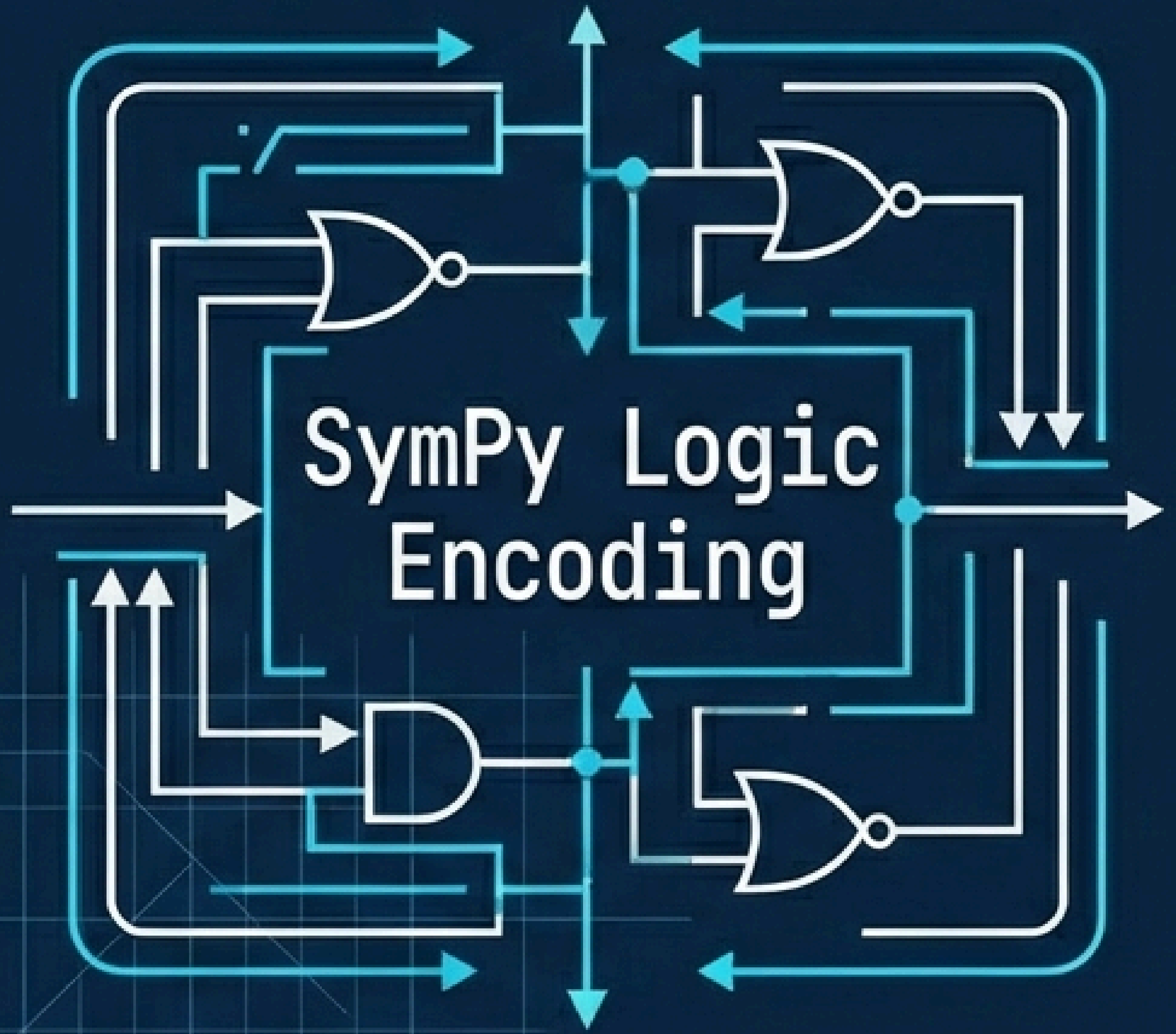
Barabasi-Albert



Critical Hub

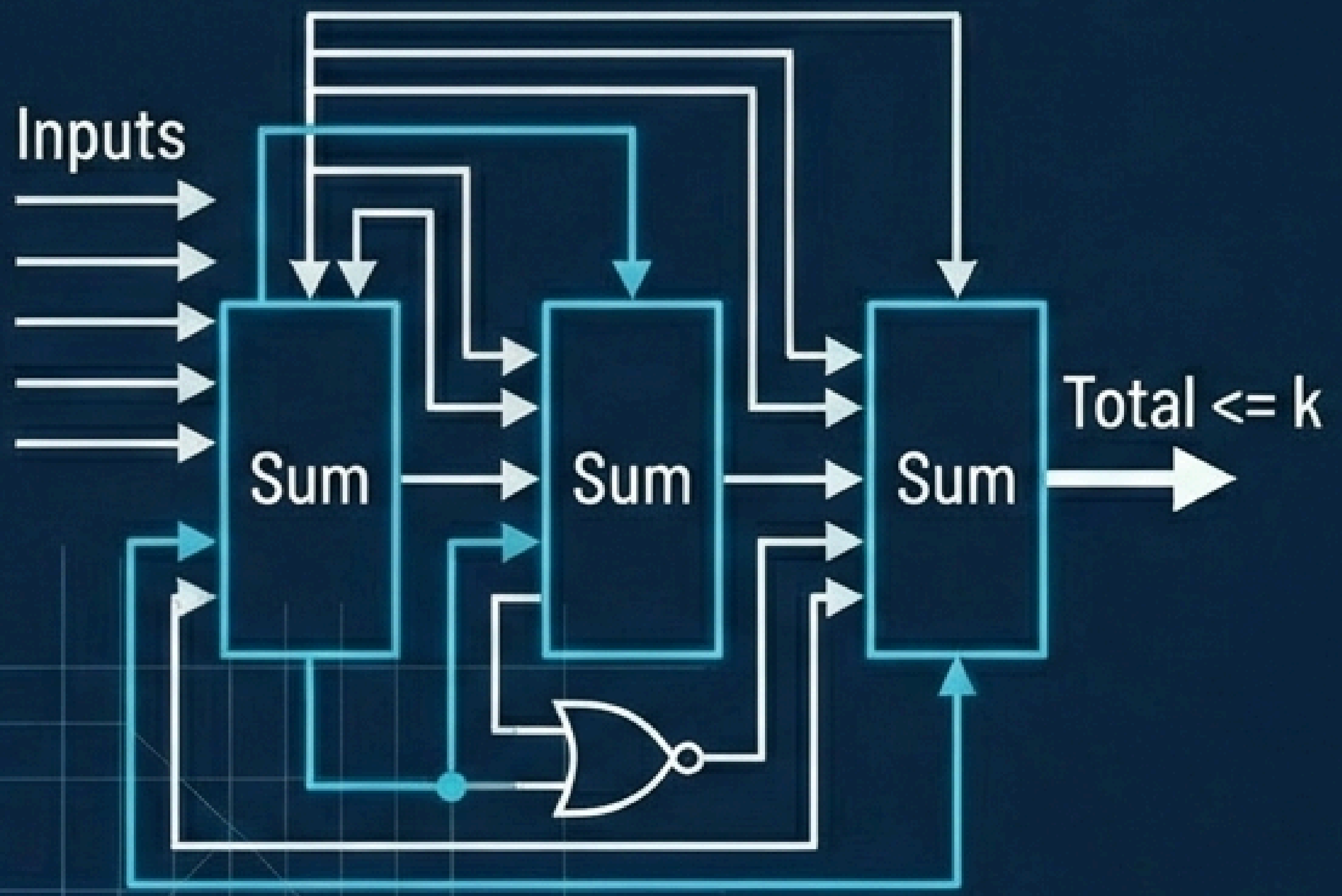
Network shape determines cost and market power.

Bonus: Exact Logic via SAT



- **Objective:** Algorithmic exploration beyond Game Theory.
- **Approach:** Boolean Satisfiability (SAT).
- **Encoding:** (x_u OR x_v) for every edge.

SAT Logic: The Cardinality Constraint



- **Challenge:** Encoding "At most k" without arithmetic.
- **Solution:** Recursive Sequential Counter Circuit.
- **Optimization:** Binary Search on k.

Conclusion & Key Insights

- **Integration:** From Selfish Agents to Boolean Logic.
- **Result:** Game Theory heuristics approached SAT Ground Truth.
- **Takeaway:** Topology is the primary determinant of security cost.

Thank You

Questions?